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## Supplementary Validation: Locating resolution-dependent behavioral correspondence in qualitative DNA-damage response models

Carlos Ramirez Ovalle  
*Pontificia Universidad Javeriana Cali, Colombia*

This supplement retains implementation-verification and exploratory-data details supporting the main biological comparison. It includes the death-receptor execution audit in Section S5. It is not independent wet-lab validation of either biological case.

### S1. Formal and software verification

#### S1.1. *Construction-level software benchmark and comparator study*

Six deterministic pairs were specified before any biological application. They provide behavioral-class ground truth by construction and test whether the software recovers intended formal distinctions; they do not provide biological ground truth. A 12-step labeled workflow supplies identity, controlled silent subdivision, label mismatch, label-order swap, and an added observable branch. The sixth pair is  $a.(b+c)$  versus  $a.b + a.c$  from Example 1 of the main article. Table S1 gives construction-level and computed classes.

On this construction-level decision task, weak bisimulation recovered all six binary weak-equivalence decisions. Truncated trace equality recovered 5/6 but accepted the trace-equivalent branching pair; LTS-GDA recovered 4/6, the structural profile 3/6, and direct PN-GDDA 2/6. PN-GDDA assigned 1.0 to identity, silent refinement, label mismatch, and order swap, while added-branch and branching pairs scored 0.9676 and 0.9657. No threshold over the six PN-GDDA scores exceeded 4/6 because two negative controls tied the positive controls at 1.0. Figure S1 reports all predeclared decisions.

The structural profile averages state-count, edge-count, label-multiset, and out-degree-multiset similarities. The trace diagnostic declares a depth-8 match when  $d_8 = 0$ . PN-GDDA was implemented from its published definition using 151 directed bipartite graphlets and the 592 published/Holmes orbit slots.<sup>1</sup> For orbit  $j$ ,

$$N_G^j(k) = \frac{D_G^j(k)/k}{\sum_{r>0} D_G^j(r)/r}, \quad A^j(G, H) = 1 - \frac{\left[ \sum_k (N_G^j(k) - N_H^j(k))^2 \right]^{1/2}}{\sqrt{2}}, \quad (\text{S1.1})$$

Table S1. Controlled benchmark; the expected class is not used by the classifier.

| Case              | Change                        | Expected               | Observed               |
|-------------------|-------------------------------|------------------------|------------------------|
| Identity          | exact copy                    | strong equivalence     | strong equivalence     |
| Silent refinement | insert internal steps         | weak equivalence       | weak equivalence       |
| Label mismatch    | replace an observable         | not comparable         | not comparable         |
| Order swap        | preserve counts, change order | not comparable         | not comparable         |
| Added branch      | add an outcome                | reference in candidate | reference in candidate |
| Early/late choice | move decision point           | candidate in reference | candidate in reference |

and PN-GDDA is the mean  $A^j$ . Synthetic LTSs use an exact state-machine Petri net encoding; curated pairs use their native nets. The 0.9 threshold is a fixed diagnostic decision, not an equivalence relation. LTS-GDA instead combines four reachable-graph orbits with labeled edge, two-step, divergence, and convergence motifs.

The catalog audit recovered 151 graphlets and 592 slots from hash-verified Holmes 1.1.1/2.0.1.2 JARs. A direction- and type-preserving automorphism partition has 576 distinct orbits; using it is reported only as sensitivity. A Java probe matched every inspected Holmes 1.1.1 topology/root assignment. On a four-node test net and its one-arc deletion, Python and Holmes scores were 0.9872027465870733 and 0.987202746587073, agreeing to 12 decimal places. Holmes is an independent software reference for the structural score.<sup>2</sup>

### S1.2. *Independent formal oracles*

Each analyzed LTS was exported in Aldebaran AUT format and evaluated with `ltscompare` from hash-pinned mCRL2 202607.0.<sup>3</sup> The tool independently checked strong bisimilarity, weak bisimilarity, and exact weak-trace equivalence with `tau` hidden. Its interface does not expose a general weak-simulation preorder. For the tau-free Example 1 and middle strictness witness only, the audit additionally uses its strong-simulation preorder, which coincides with weak simulation when neither input has silent transitions; both directions agree. No such weak-preorder claim is made for systems containing silent transitions.

A separately coded attacker–defender game grows the least set of positions from which an unmatched move can be forced. It agreed with the production simulation algorithm on all 67,600 ordered pairs among all 260 one- and two-state LTSs over  $\{a, \tau\}$ . This is exhaustive within that finite universe, not a proof for arbitrary systems. Across the six synthetic pairs, six public-model controls, and nine HPN comparisons below, Python and mCRL2 agreed on all 42 reported strong/weak decisions. Theorem 1 of the main article, finite exhaustive testing, and external-tool comparison are distinct evidence layers.

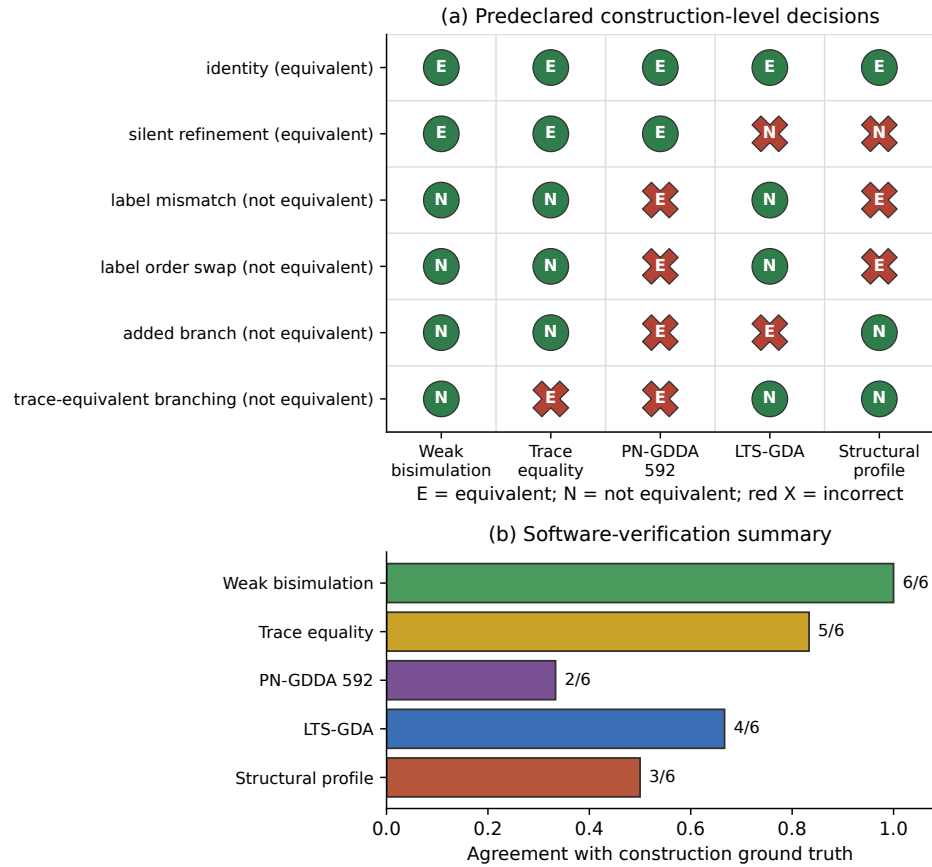
**Controlled benchmark: software verification only**

Fig. S1. Controlled software/formal benchmark. (a) Decisions for six cases whose labels were fixed by construction; a red cross marks disagreement with that label. (b) Agreement count on this limited task. Trace equality loses branching; PN-GDDA is label-blind; LTS-GDA retains local labels but does not abstract silent refinement or recursively match successors. The panel verifies implementation behavior and does not measure biological validity or generic method superiority.

**S1.3. Observed computational scaling and reproducibility**

Linear workflows of 8, 16, 32, 64, 96, and 128 transitions were compared with an identity and a silently refined copy. Median runtime over three repetitions rose from about 0.4 ms for nine states to about 0.6 s for the largest pairs. The 128-step identity relation had 16,641 candidate pairs; refinement increased it to 18,705 pairs over 129 and 145 states. Every class remained as constructed (the complete plot and table are archived in the repository). Timings used Python 3.11.8 on an Apple M3 Pro

Table S2. Public GINsim models and control outcomes. S/W/T denote strong, weak, and exact weak-trace equivalence.

| Model                | Format | States | Edges | Copy  | Silent | Perturbed |
|----------------------|--------|--------|-------|-------|--------|-----------|
| Mammalian cell cycle | SBML-q | 826    | 3,413 | Y/Y/Y | N/Y/Y  | N/N/N     |
| p53-Mdm2 damage      | GINML  | 17     | 26    | Y/Y/Y | N/Y/Y  | N/N/N     |

with 18 GiB memory and are machine-dependent.

The Python standard library implements reachability, formal relations, parsers, and controlled models. NumPy, pandas, Matplotlib, NetworkX, pydot, and Graphviz support tabulation and figures. Fixed seeds, source hashes, machine-readable outputs, unit tests, an executed notebook, environment files, and continuous integration are public. Runtime CSV values are excluded from byte-level regression because they depend on hardware; all scientific outcomes and other result tables are deterministic.

## S2. Cross-format ingestion and execution controls

Two externally authored GINsim models test the complete import-to-comparison path: the 13-component mammalian cell cycle in SBML-qual and the four-component multilevel p53-Mdm2 model in GINML.<sup>4-7</sup> Files, URLs, and SHA-256 values are archived. Unitary-asynchronous updates move one enabled component by one level and label its direction. Each base model was compared with an exact copy, five silent subdivisions, and a reachable-label perturbation. All six controls recovered their predeclared strong/weak/exact-trace outcomes in Python and mCRL2 (Table S2).

Synchronous execution reached 13 of 826 asynchronous cell-cycle states (Jaccard 0.0157) and eight of 17 p53-Mdm2 states (Jaccard 0.471), confirming sensitivity to update semantics. These are cross-format execution controls, not biological validation.

## S3. Exploratory HPN-DREAM/CASPOTS held-out analysis

The public HPN-DREAM phosphoproteomic series and CASPOTS Boolean families contain 72 BT20, 191 BT549, and 21 MCF7 networks.<sup>8,9</sup> One representative per cell line was selected before test responses were read: the model with maximum mean within-family Jaccard similarity between active clauses, with exact lexicographic tie-breaking. This structure-only rule selected zero-based rows 24, 137, and 9 (Table S3); an anti-leakage test forbids access to test files during selection.

The interface is the intersection of seven measured PI3K/MAPK/mTOR readouts. The three common test perturbations are mTOR inhibition, IGF1 plus mTOR inhibition, and 4EBP1 plus mTOR inhibition. Measured time-zero values initialize readouts; stimuli and inhibitors are clamped; other relevant nodes start at zero. Primary asynchronous LTSs label interface updates and hide all others. The nine

Table S3. HPN-DREAM families and structure-only representatives.

| Cell line | Family | Medoid row <sup>a</sup> | Clauses | Mean Jaccard |
|-----------|--------|-------------------------|---------|--------------|
| BT20      | 72     | 24                      | 29      | 0.815        |
| BT549     | 191    | 137                     | 25      | 0.854        |
| MCF7      | 21     | 9                       | 23      | 0.851        |

Note: <sup>a</sup>Zero-based row; selected from model structure only.

cross-cell pair-condition comparisons were repeated synchronously; this stress test is not a most-permissive analysis. CASPOTS itself evaluates asynchronous trajectories.<sup>9,10</sup>

Experimental distance is RMSE over common post-zero times and readouts. Spearman associations use an exact one-sided permutation of the three cell-pair labels within each condition ( $3!^3 = 216$ ). CASPOTS/Clingo compatibility scores consume optimization to completion, use deterministic secondary tie-breaks, and were reproduced twice. Native specificity uses the exact sign permutation of three native-minus-mean-cross differences.

Selected LTSs had 2–44 states. None of the nine cross-cell pairs was weakly bisimilar: six had one-way simulation and three were not comparable. Direction was retained explicitly, with three left-model-in-right-model and three right-model-in-left-model relations. These categories do not form a justified one-dimensional order, and the held-out RMSE is symmetric between each cell pair. We therefore removed the former ordinal class coding and performed no scalar formal-class test. Descriptively, median RMSE was 0.2418 for the six one-way relations and 0.2071 for the three non-comparable pairs; after preserving direction, the corresponding medians were 0.2757, 0.1919, and 0.2071, each based on only three observations. These summaries are not inferential evidence.

The two predeclared continuous diagnostics were retained. Truncated-trace distance gave  $\rho = -0.596$  (exact one-sided  $p = 0.556$ ), and LTS-GDA distance gave  $\rho = 0.669$  ( $p = 0.097$ ) under 216 condition-stratified permutations. Neither association is conclusive. Synchronous updating preserved seven of nine formal classes.

Native medoid RMSE was 0.33024 (BT20), 0.31335 (BT549), and 0.27726 (MCF7), against discretization RMSE 0.32944, 0.30079, and 0.27726. Native excess-RMSE advantage over mean cross-cell medoids was 0.00212 and not established by the exact sign test ( $p = 0.25$ ; Fig. S2). The result demonstrates an exploratory held-out compatibility check and separates formal similarity from empirical response similarity; it does not demonstrate cell-line specificity or predictive discrimination.

#### S4. Quantifier and direction audit

The production implementation was not changed to conform to a proposed reinterpretation of simulation. Instead, the audit independently constructs silent closures

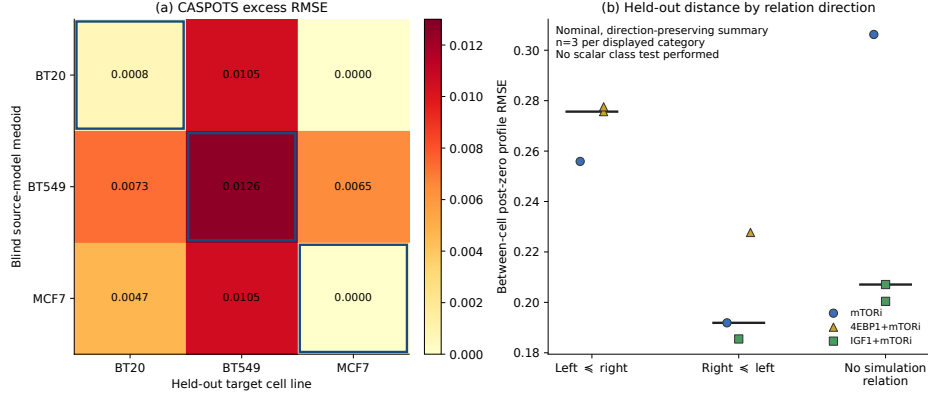
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Fig. S2. Exploratory held-out HPN-DREAM analysis. (a) CASPOTS excess RMSE by source medoid and target; outlined cells are native contexts. (b) Between-cell profile RMSE with the direction of weak simulation retained. Horizontal bars are medians; each category has  $n = 3$ , and no scalar formal-class test was performed. Neither panel establishes native cell-line specificity.

and weak targets from raw edges, and grows a least losing set in simultaneous attacker-defender rounds. It does not call the production adjacency, closure, weak-step, or relation routines. Exact weak-trace inclusion is checked by a finite epsilon-NFA subset-product traversal, treating all original states as accepting, without a depth cutoff.

For Example 1 in the main article, complete acyclic path enumeration also gives  $\{\epsilon, a, ab, ac\}$  on both sides. The supplied five-pair early-to-late relation satisfies every transfer clause. The reverse calculation first removes  $(\ell_1, e_b)$  because  $c$  cannot be matched and  $(\ell_1, e_c)$  because  $b$  cannot be matched; the initial pair subsequently has no surviving reply. The machine-readable deletion certificate records these dependencies.

All three strictness witnesses and 32 ordered model pairs (six synthetic, three GIM interfaces, five curated modules, and 18 asynchronous/synchronous HPN comparisons) agree on strong/weak bisimilarity and both weak-simulation directions between the production code and independent audit. Every synthetic class still agrees with its construction-level expectation. Direction fields explicitly state `left_simulated_by_right` and `right_simulated_by_left`.

The complete one-/two-state universe contains 260 named LTSs over  $\{a, \tau\}$  with initial state zero, hence 67,600 ordered pairs. The expanded audit found no simulation or weak-bisimulation disagreement and no failure of the hierarchy or trace-inclusion implications. Simulation is reflexive and transitive on this universe. There are 41,080 weakly bisimilar ordered pairs and 4,848 mutually simulated but non-bisimilar ordered pairs. No one-way simulation with exact trace equality occurs in this small universe; the larger acyclic Example 1 supplies that witness. The finite enumeration supports software correctness only; it does not replace the general

proofs.

Reproduce the new audit using `python -m src.formal.audit` and `python -m unittest tests.test_formal.audit -v`. The four JSON/CSV reports in `results/` include the Example 1 certificate, hierarchy witnesses, all model-pair directions, and exhaustive counts. The main article’s appendix retains all definitions needed for its claims, the relation-deletion proof, and explicit proofs of the three failed converses. Detailed Petri nets remain in the main article rather than being removed during compression of validation material.

## S5. Death-receptor execution and witness checks

### S5.1. *Provenance, initialization and reduction*

The source is the original Calzone 2010 GINsim file `Calzone_Cell_Fate.zginml`, not the later multiscale adaptation. Its SHA-256 is

```
1cdbf6a3fb00fcf7ce9d63cb753a9bf4a58
ea1e16486abb6a49a2016634a6993.
```

The stored initial-state records omit input assignments. We explicitly set `TNF=FADD=ATP=cIAP=1` and all remaining nodes to zero. DR-FB- changes only the read of CASP3 within the CASP8 rule, preserving CASP3’s own rule and every other component. All 16 combinations of the four CASP8 regulators are tested against the declared original and edge-deletion truth tables.

The original prospective configuration and two dated resource amendments are retained without overwriting their history. The first object-based execution exceeded 200,000 states; the hidden-sink quotient also exceeded that limit and a diagnostic one-million-state limit. These runs produced no formal relation decision. Compact Boolean enumeration then completed all four reachable graphs. The 28-node source has three fixed inputs in the sustained protocol and three omitted hidden sinks; withdrawal makes TNF dynamic. The retained universe is therefore bounded by  $2^{22}$  and  $2^{23}$  Boolean valuations, respectively, before reachability restriction.

Removal of NonACD, Apoptosis and Survival is not the source paper’s reduced conceptual model. These three output nodes regulate no retained component and are not in the fixed interface. Relating states with identical retained coordinates gives a weak bisimulation: output updates are matched by zero silent steps, and retained updates have identical enabling conditions and successors in both directions. Output values can independently settle at a terminal retained state. Terminal classification uses NFkB/CASP3/MPT and ATP, not the omitted delayed output copies. Strong relations are not inferred from this weak reduction. The selected full 28-node continuations are also checked directly against their quotients and reproduce the reported weak relations.

**S5.2. *Exact checks, limits and distinct comparison scopes***

The sustained graphs contain 1,044,457 and 547,021 states, with 10,296,917 and 5,062,632 edges. The withdrawal graphs contain 3,133,393 and 2,140,569 states, with 32,980,090 and 21,544,296 edges. All reachable terminal components are singleton fixed points. This is enumeration of possibilities, not a reproduction of Monte Carlo phenotype probabilities or physical-time calibration in the original publication.

The exact epsilon-NFA subset-product search accepts every nonempty state subset and rejects only the empty subset, thereby checking prefix-closed finite weak languages without a depth cutoff. The original limits of 10,000 subset pairs, 250,000,000 stored bytes and 180 seconds did not decide the sustained comparison. A subsequent resource amendment increased only its search limits to 100,000 pairs, 2,000,000,000 bytes and 600 seconds. It reached 6,158 pairs and 1,999,975,480 stored bytes before the memory cap, after finding this DR-FB+-only 12-action word:

MPT\_up, CASP3\_up, CASP3\_down,  
 NFkB\_up, NFkB\_down, MPT\_down,  
 CASP3\_up, CASP3\_down, CASP3\_up,  
 CASP3\_down, CASP3\_up, CASP3\_down.

The complete subset search remains unfinished, but this finite counterexample refutes global sustained trace equality. The reverse inclusion remains undecided. The reproducible driver rechecks the word explicitly, even if a later discovery search hits a resource cap sooner.

Independent SciPy directed reachability gives 8,264, 7,936 and 8,064 accepting states for the last three prefixes in DR-FB+, versus 744, 120 and zero in DR-FB-. A reconstructed 53-update positive path satisfies the original 28-node rule callbacks. Independent mCRL2 antichain checks include the word's finite prefix-language in DR-FB+ but not in DR-FB-, using a 180-second limit per command. The negative sparse check uses the proven hidden-sink projection; it is not a direct exhaustive search of the full 28-node product. The latter reached a 100,000-state cap for both variants and remains inconclusive. These distinctions prevent a partial audit from being presented as a proof.

The failed sustained inclusion also refutes the corresponding weak simulation and weak bisimulation, without completing a global relation game. Direct global games exceed their one-million-pair budget. Earlier full-graph external checks had 180-second limits, and an additional global antichain inclusion check reached 600 seconds without a decision. None of these timeouts is the basis of the inequality result. Withdrawal search retains its original limits.

The withdrawal search instead finds exact counterexamples in both directions:

DR-FB+ only: withdraw\_TNF, CASP3\_up, CASP3\_down, CASP3\_up.  
 DR-FB- only: CASP3\_up, withdraw\_TNF, CASP3\_down.

For each word, epsilon-closure propagation leaves a nonempty accepting subset in one system and an empty subset in the other. Once both inclusions are refuted,

exhaustive exploration of every other word is unnecessary. Weak simulation would imply the corresponding trace inclusion, so both global weak simulations, and therefore both bisimilarities, fail for the intervention protocol. This inference is from exact counterexamples, not from a timeout or from the state-specific continuation certificate.

A second implementation checks these two words using a direct product of the original 28-node tuple states and word positions. It does not use compact truth tables, projected graphs, or the sink reduction. For the DR-FB+-only word it finds an accepting execution in DR-FB+ (5,356 visited product states) and exhausts DR-FB- (3,747 states) without acceptance. For the DR-FB-only word it exhausts DR-FB+ (929 states) and finds acceptance in DR-FB- (1,124 states). Thus both global counterexamples are independently confirmed in the full source-rule models.

The automatically selected shared state occurs after eight raw updates. Its sustained continuations are three-state/two-edge silent graphs, with exact language  $\{\epsilon\}$ . Allowing withdrawal yields 12-state/17-edge and 34-state/54-edge graphs. Exact subset-product exploration has four reachable subset pairs. The positive continuation is simulated by the negative one; the reverse fails. The deletion certificate begins at their initial pair and retains every possible weak defending reply, with strictly smaller deletion ranks at dependencies. Independent verification reconstructs these replies from the raw edges, and rejects a deliberately truncated certificate in the unit tests. mCRL2 agrees on strong/weak bisimilarity and exact weak traces for both sustained and intervention-conditioned comparisons. Its arbitrary weak-simulation preorder is not assumed to be available.

### S5.3. History aggregation and the event-depth scan

The common raw path is TNFR, DISC\_TNF, CASP8, BAX, MOMP, Cyt\_c, apoptosome, CASP3 activation, in that order. Its visible projection contains only CASP3 activation. Exact closure propagation shows 78 hidden states in each model compatible with that visible history. Their aggregated terminal futures after immediate withdrawal are apoptosis/naive/necrosis for DR-FB+ and naive/necrosis for DR-FB-. The selected retained state is therefore not claimed to be uniquely identified by the visible prefix.

The scan propagates all states reachable after exactly  $k$  raw component updates, for  $k = 0, \dots, 40$ . It then applies the same single TNF withdrawal and reads complete terminal-future sets from the fully enumerated controlled graph. There is no future-depth truncation. Terminal states are not padded with fictitious updates. At depth eight, one of 73 states in DR-FB+ has both unique apoptotic terminal fate and persistent CASP3 after withdrawal. No earlier depth has such a state. The reported counts weight states equally only as counts, not as trajectories or cell probabilities. The scan does not assert that no later behavior exists beyond depth 40; the entire reachable graph and the selected witness are analyzed separately.

The naive signature is a logical readout combination, not evidence of cellular

rescue after irreversible apoptosis. A joint marker assay after withdrawal would test a model-derived prediction; cell type, physical timing, selective edge manipulation and wet-lab confirmation remain outside this study.

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